RA genetic and macrophage-state evidence across public multi-omics resources

GeNeDis 2026 · Xia Wu, PhD student, University of Iowa · xwu76@uiowa.edu

Summary

This workflow links complementary RA data layers and tests where candidate signatures fail. We reanalyse a published synovial RNA/ATAC atlas, map a sorted-macrophage discovery signature, compare bulk cohorts and assess spatial, protein and genetic evidence. CD4 genetic enrichment and IFN-associated macrophage features in tissue occur together. These data do not establish a directional causal chain from lymphocytes to macrophages.

Our contribution is a reproducible evidence map across resources, with integration and model baselines, clear score definitions and negative controls. The atlas states and their original genetic enrichment come from Gupta et al., Nature Genetics 2023. We keep the authors' fine-state annotations, including M-6 STAT1+ CXCL10+ macrophages. We do not present these states as new discoveries.

Results

Atlas and integration (Fig. 1)

The joint RNA/ATAC analysis contains 31,520 nuclei from 12 donors (11 RA, 1 OA). The author-defined M-6 state contains 65 nuclei. The benchmark compares RNA, ATAC, concatenation, WNN and one MOFA+ configuration. It uses shared cells and labels derived from both RNA and ATAC (Fig. 1e). Concatenation and WNN score best overall, but no method isolates M-6 as a cluster. Rare-state neighbourhood purity is highest with RNA alone. This evaluates only the tested configuration. It is not a general ranking of integration methods.

Genetic association and tissue niches (Fig. 2)

Conditional S-LDSC links broad T-cell chromatin to RA heritability in Ishigaki EUR and FinnGen (tau z about 4.7). CD4 Tph/Treg/OX40-associated states are among the leading subtypes. T2D is a trait control. scDRS uses expression and the same GWAS traits. Its agreement with S-LDSC across states is modest (rho 0.26, P 0.047; Fig. 2d), and it is complementary methodological evidence. It has not been established that all GWAS participants are independent across resources.

Fine-mapping supports an ANKRD55-region variant across ancestries (Fig. 2e). This does not extend to all T-cell enrichment, because the East Asian analysis shows no broad T-cell enrichment. A non-positive myeloid conditional coefficient cannot establish that macrophages are downstream. M-6 is too small for its own S-LDSC annotation.

CXCL13-positive T cells have more B-cell neighbours in all four RA 5K sections (Fig. 2f–h). Each section reaches the empirical permutation resolution of about 0.001. Fig. 2i reports empirical P and does not extrapolate large z scores into extreme significance. Global label shuffling keeps cell positions but not label clustering or tissue compartments. This spatial association does not show that CXCL13 created the aggregates. CXCL13/CXCR5 communication ranks point to a compatible interaction for follow-up.

Distinct macrophage measurements (Fig. 3)

S1: sorted-cell discovery signature. Fifteen genes are higher in active than in remission sorted macrophages at FDR <0.05. They include metallothioneins, glycolytic genes, APOL6 and GBP5 (Fig. 3a). Thirteen of these genes are measurable in the multiome comparison. S1 scores highest in M-6, and donor RNA and ATAC gene-activity scores agree (rho 0.84; random-set P 0.008; Fig. 3b–c). Both modalities measure the same nuclei. No S1 gene has an FDR-significant peak-gene link in this analysis. The independent multiome association with donor inflammation is not significant (10 RA donors, rho 0.43, P 0.21). In bulk RA/OA tissue, S1 fails its association and matched-random specificity tests (P 0.327; random-set P 0.525). When discovery donor SA139 is dropped, only 1/15 genes stays significant, although the positive directions remain. S1 remains exploratory.

S2: predefined IFN programme. CXCL9, CXCL10, CXCL11, STAT1, GBP1, GBP5, IDO1, IRF1, CD274 and APOL6 form a separate ten-gene score. S1 and S2 share only APOL6 and GBP5. S2 is associated with IFNG-positive lymphocyte neighbourhoods in 3/4 RA 5K sections (Fig. 3e). It is also associated with more IFN-high Visium spots in RA (Fig. 3f). In bulk tissue, its RA/OA difference goes beyond matched random sets (random-set P 0.002; P 0.001 after adjustment for macrophage abundance; Fig. 3h). These are S2 associations, not a successful S1 bulk validation.

The limits of global spatial shuffling also apply here.

Hallmark pathway/protein evidence. Disease and treatment pathway NES values are inversely related (Fig. 3d). The protein panel uses Hallmark IFN-gamma response genes, not the exact S1/S2 set. RA and healthy donor scores differ in both lining and sublining (6 versus 3 donors per compartment, Mann-Whitney P 0.024 each; Fig. 3g). Fibrous and immune sublining regions are averaged within each donor before testing and plotting. Matched random protein sets support specificity in lining (0/1,000 as extreme; resolution about 0.001) but not in sublining (P 0.339). Xenium, Visium and proteomics share blocks and cohort, so they are not three independent patient replications.

Treatment. In the anti-TNF cohort, 13/15 S1 genes are measured. The technically residualised S1 score decreases within good responders (median change -0.70, paired P 0.014; Fig. 3i). However, the change does not differ significantly between good responders and nonresponders (P 0.131). This does not establish response specificity or prediction. Apparent baseline prediction is confounded by sequencing depth.

Candidate regulators and targets (Fig. 4)

RNA regulon and motif comparisons point to NF-kB, STAT1 and IRF-related candidates (Fig. 4a–b). The M-6-bearing units come from two donors. The candidate flag combines BH-adjusted gene-label-permutation q <0.05 with a top-decile motif ranking. Gene permutations do not give donor replication or causal evidence.

Two IRF5 GWAS signals colocalise with myeloid expression QTLs (Fig. 4c). Under the model assumptions, this supports shared association signals. Single-instrument Wald ratios link higher CTLA4 expression to lower risk and higher IL6ST expression to higher risk (Fig. 4d). Pleiotropy cannot be tested. These estimates do not recover drug efficacy. Abatacept binds CD80/CD86 and tocilizumab binds IL6R, while IL6ST encodes gp130. IL6ST and ANKRD55 share an instrument and cannot be separated here. IRF5 directions conflict across datasets and signals. Drug annotations give biological context only.

The transparent 11-layer gene ranking (Fig. 4f–g) sets priorities among follow-up hypotheses. The layers overlap, and the scores are not causal probabilities. A low Open Targets association describes that database release only. It does not prove a lack of literature or guarantee novelty.

Extension and boundaries (Extended Data Figs. 1–2)

The extensions use separate 30-gene M-6/Tph programmes derived from the author-defined atlas states. In synovial fluid (9 paired patients), the M-6-like fraction decreases in 7/9 (P 0.027). The paired change in M-6 pseudobulk score is not significant (P 0.129), even though the matched-random comparison gives P 0.005. Cluster evidence is borderline (P 0.055), response labels are not available, and two treatments are pooled.

In LuCA, macrophage M-6 scores are higher in tumour than in adjacent lung in 20/29 paired donors (P 0.0038). The macrophage/CD4 Tph correlation is modest and not conventionally significant (rho 0.24, P 0.092), although the random-set P is 0.030. The TCGA primary survival tests adjusted for age and stage are negative. The leukocyte-adjusted LUAD result is a sensitivity analysis. It can be affected by tumour-cell expression and by multiple testing. These are cross-disease associations. They do not show a shared treatment mechanism.

Classifier evaluation uses training-only repeated nested CV (25 outer folds) and permutations that repeat the same procedure. In RA versus OA (174 biopsies), the eight-programme model has a mean AUC of 0.900, against 0.979 for the genome-wide baseline. The approximate corrected resampled comparison favours the genome-wide model (P=0.00596). In ACPA-positive versus ACPA-negative RA (139 biopsies), the matching means are 0.797 and 0.726. The approximate corrected comparison finds no significant difference (P=0.149). The prioritised-gene model does not do significantly better than 100 random gene sets matched for feature count, in either task (empirical P=0.2475 and 0.5149). Both programme-model label tests reach P=0.005, the smallest value possible with 199 permutations. This shows internal association under this procedure, not external prediction or biological specificity. Full results are in QC report section 33 and classifier_baselines.tsv. These CV and permutation estimates are conditional on the given feature lists. They do not correct for earlier exploratory choices of tasks or programmes. These are exploratory tests for each task. The SD across correlated folds is not a confidence interval, and no independent clinical model validation is available. Blood transfer, deconvolution sanity, L1000 specificity and chromatin-before-RNA timing fail or remain uninterpretable. They are kept in Extended Data Fig. 1.

Final classifier and validation records

See classifier_baselines.tsv and classifier_protocol.json for all estimates and the resampling protocol.

Revision and validation record (GeNeDis 2026)

Participant: Xia Wu, PhD student, University of Iowa · xwu76@uiowa.edu. Scope: an exploratory multi-omics integration workflow that follows the official public deliverable list. No separate organiser template was supplied.

Revision traceability

Issue	Correction	Evidence
Protein plot and test used different sublining groups	All immune/fibrous/sublining regions are pooled within each donor before plotting and testing	Fig3g source values; sample/P assertions; submission checks
Hidden figure cache	Steps 07/20 export small plotting inputs, and the six figures render from packaged tables	Figures rebuilt in isolation
Classifier preprocessing and null mismatch	Every fitted transform stays inside the inner/outer training folds. The 199 null labels repeat the same tuning/CV. The 100 random controls match the retained F2 gene count	Protocol JSON, fold/null tables, invariance tests
Nominal TF selection	BH adjustment over all 464 RNA TF tests; the nominal flag is kept for transparency	Corrected q values and candidate flags; IRF5 is not a corrected RNA candidate
Zero positive-neighbour exposure	The healthy Xenium section has no IFNG-positive exposure, so its coefficient and P are marked non-estimable, not zero/null	Extra step-20 rerun; test_status and submission assertion
Extreme spatial combined P	Normal-tail extrapolation from effect z removed; section-level effects and empirical P kept	Xenium/Tph tables; figure panel 2i
Mixed signature claims	Clear S1/S2/Hallmark/M-6/Tph definitions and a score registry	Narrative, figure titles/legends and evidence_map.tsv
Causal/drug overclaims	Association/candidate wording; molecular distinctions for CTLA4-Ig and IL6ST/IL6R; two-donor and pleiotropy limits stated	README, narrative, QC report, presentation and email
Independence/novelty	Published states and original genetic findings credited; same-nucleus and shared-block measurements marked as such; GWAS independence not claimed	Sources and narrative
Knowledge-graph semantics	S2 has its own score node; spatial nodes reflect measured cells; edges describe associations or predicted interactions	Graph source, nodes/edges and semantic checks
Reference APIs could change	Saved copies of Open Targets 26.09 and an API/GMT/eQTL snapshot, with per-file checksums and a verified restore	Reference manifest and restore log
Platform/conversion dependencies	Python locks, a documented R inventory and conversion, Rosetta/MAGMA constraints, pinned tool hashes	Environments, R conversion comparison and tool checks
Partial S-LDSC output could silently shrink the FDR family	The step stops on any failed/missing model and needs all 195 unique finite results before BH	Completeness failure-mode tests; final step-33 entry-point check and submission checks
Old captions/slides/contact fields	Panel notes, the six-figure set and the slide deck rebuilt; individual details filled in	Final artifact checks
Submission package could change itself during checking	Quick verification runs in a temporary project and rechecks the original hashes	Quick verification of the extracted package
Report layout and wording	The report and slides use black and grey headings, text, links and table lines. The README, report, figure legends and slides were reworded in plain English. Numbers, tables and figures did not change	Rebuilt report and slides; number and name check of every edited text

Validation status

Ordered analysis: PASS. All 43 runs completed successfully (41 numbered steps, with steps 22/32 repeated to resolve dependencies). The analysis began 2026-10-20 13:30:36, and the final figure export ended 2026-10-20 16:17:35 (America/Chicago; 2.78 h elapsed).

The project started with empty analysis intermediates and results. Primary raw files were byte-identical local copies of the public input files, checked by hashes. The API/GMT/eQTL reference snapshots were kept fixed on purpose. Independent branches were scheduled after their data dependencies. The ordered driver reused only derived files generated inside this same clean validation project. The early step-36 branch was stopped after MAGMA to control memory. A resumed step-36 branch then completed scDRS after MOFA released memory. The ordered step 36 checked and reused these results by their exact input and output hashes. These scheduling actions are recorded in the logs, and no old analysis model was substituted. During validation, the 22 baseline annotation files were converted with the pinned LDSC parser to Parquet, a format LDSC supports natively. Exact value/dtype round-trip checks were run, and the original gzip inputs were kept. Later regressions used this storage cache. Model definitions and numeric input values did not change. The conversion evidence is in logs/annotation_parquet_cache.json. An extra step-20 rerun marks the healthy section with no IFNG-positive neighbours as non-estimable. It keeps the RA estimator and the permutation RNG sequence unchanged, and its completion is recorded separately. A further step-33 run checked the final cache-preparation entry point. It reused only the regressions freshly computed in this project and is logged separately.

Table comparison against the pre-revision baseline: {"identical": 103, "expected revision": 7, "new output": 2}. Differences in the classifier, corrected TF selection and spatial summaries are expected and documented below. The full per-table comparison is in results/reproducibility_check.tsv. New null-distribution tables are marked as new outputs.

Checks completed before the final package was saved:

• Four Seurat-converted input files were reproduced with identical decompressed contents.

• Invariant checks passed for training reference, query independence, nested holdout, matched permutation and matched random-feature count.

• The first isolated quick check reproduced 110 result tables, 15 processed tables and all six figures, without the large single-cell raw/interim objects.

• Final submission consistency checks are in results/submission_checks.json.

• Verification of the extracted archive and the final internal re-review are recorded in the release verification note, after the archive is finalised.

Interpretation boundaries

M-6 has 65 discovery nuclei, and regulatory comparisons use two donors. Global spatial label shuffles do not keep tissue compartments or spatial label clustering. The independent inflammation and bulk tests of signature S1 are negative. The anti-TNF result is a within-group change, not a significant difference between response groups. Wald ratios are single-instrument, expression-associated estimates, not drug effects. All classifiers are internally validated, and no clinical utility is established. Modalities from a shared cohort give complementary measurements, not independent human replication.

Revised classifier values

Task	Model	Mean AUC	SD
RA vs OA	F0 covariates	0.574	0.143
RA vs OA	F1 eight programme scores	0.900	0.048
RA vs OA	F2 top-30 prioritised genes	0.933	0.042
RA vs OA	F3 genome-wide elastic net	0.979	0.029
RA vs OA	F4 random 30-gene sets (median of 100)	0.909	0.046
ACPA+ vs ACPA- (RA)	F0 covariates	0.332	0.079
ACPA+ vs ACPA- (RA)	F1 eight programme scores	0.797	0.062
ACPA+ vs ACPA- (RA)	F2 top-30 prioritised genes	0.647	0.083
ACPA+ vs ACPA- (RA)	F3 genome-wide elastic net	0.726	0.089
ACPA+ vs ACPA- (RA)	F4 random 30-gene sets (median of 100)	0.649	0.054

AUC SD is taken across correlated outer folds, and random-set SD across draw means. Neither is a confidence interval.

• **RA vs OA**: F1 permutation p 0.005, with the same repeated nested-CV procedure. F2 vs random empirical p 0.2475; SD across random-set mean AUCs. F1 − genome-wide: -0.079; t -3.017, p 0.00596; approximate corrected resampled test.

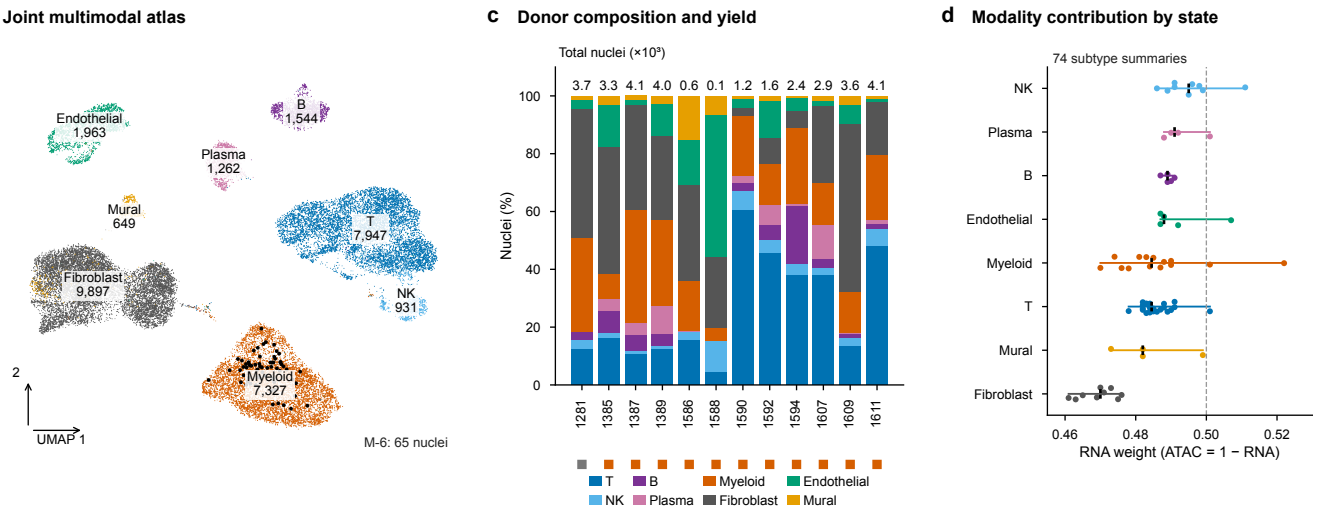
• **ACPA+ vs ACPA- (RA)**: F1 permutation p 0.005, with the same repeated nested-CV procedure. F2 vs random empirical p 0.5149; SD across random-set mean AUCs. F1 − genome-wide: 0.072; t 1.49, p 0.1492; approximate corrected resampled test.

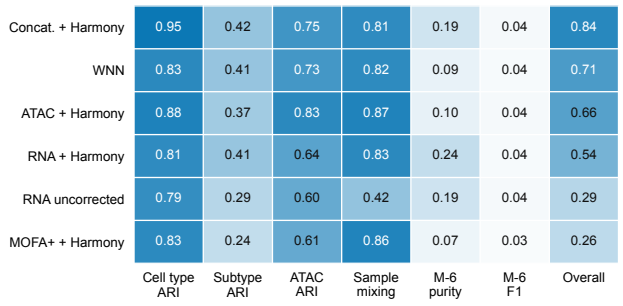
All estimates are internal to GSE89408. A low label-permutation P shows association under this modelling procedure. It does not show external predictive performance or specificity of the candidate biology.

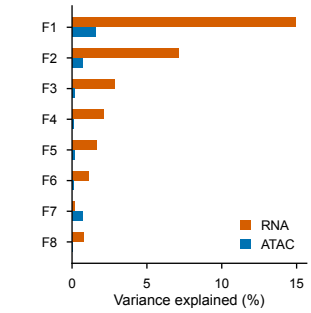
Fig1 | Study design, published atlas states and integration baselines

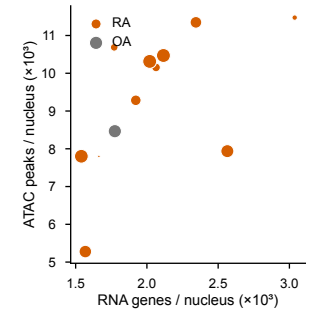
a Study cohorts and evidence layers

Genetics	Joint RNA + ATAC	Macrophages	Bulk synovium	Spatial tissue	Protein / context
3 RA GWAS	31,520 nuclei	26 donors	218 biopsies	17 Xenium sections	24 LCM regions
EUR · EAS · FinnGen	12 donors	Healthy · active · remission	46 paired anti-TNF patients	12 Visium sections	Blood · lung cancer
S-LDSC · PIP · QTL · MR	GSE243917	E-MTAB-8322	GSE89408 · GSE198520	Matched tissue blocks	Targets · perturbations









a Datasets used in the study. Counts come from the existing study description.

b WNN coordinates from the reanalysis. Colours and labelled counts show eight broad cell types. Black points mark M-6.

c Cell-type make-up of the nuclei in each donor, with the yield printed above each bar. Squares below mark RA (vermilion) and OA (grey).

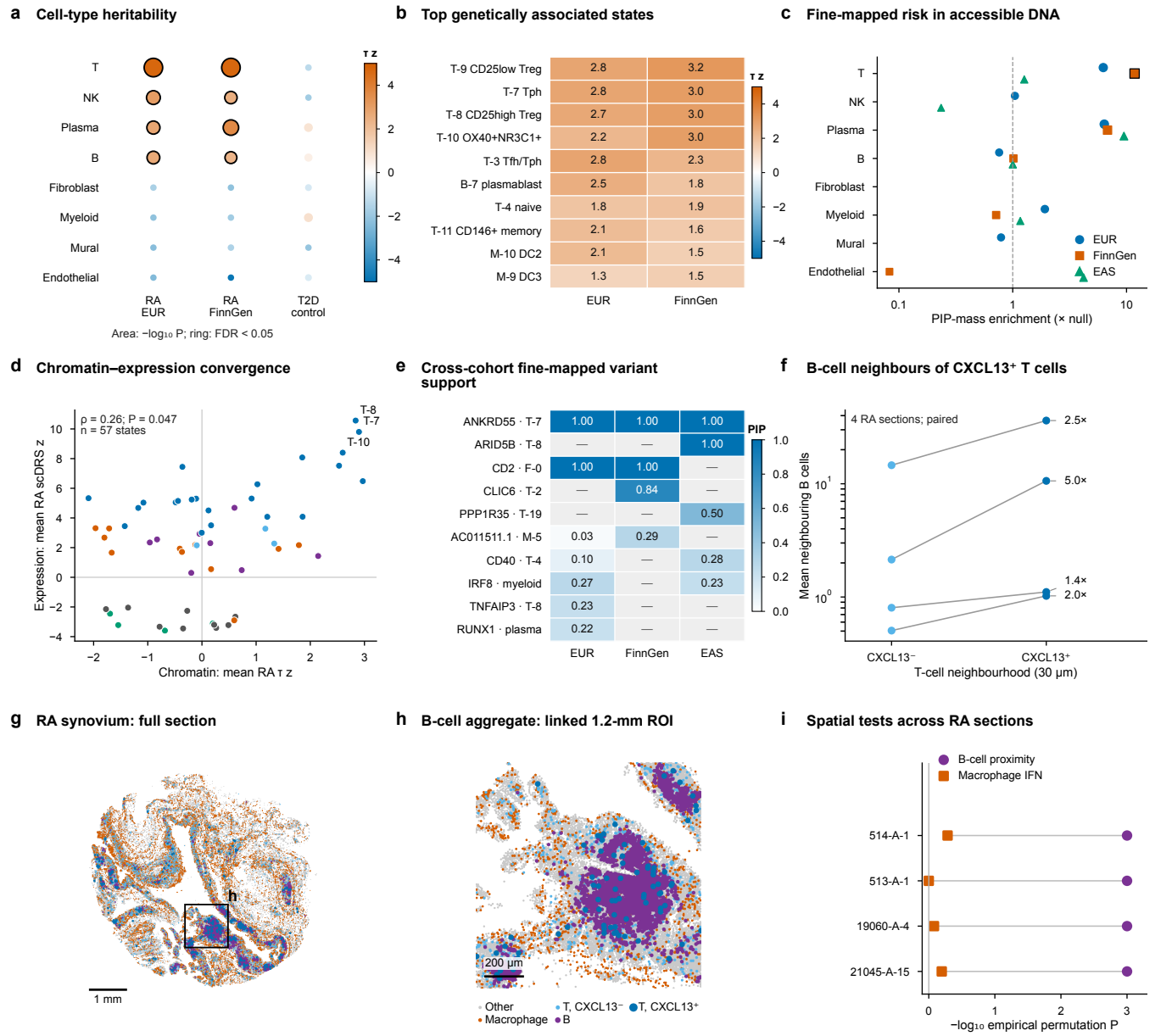
d RNA weights by subtype. Points are subtype summaries, black ticks are cell-type medians and lines run from the lowest to the highest subtype value.

e Raw benchmark metrics on a 0–1 scale. All columns use the same colour scale (white = 0, saturated blue = 1). The overall score is taken as given by the analysis.

f Percentage of variance explained by each of the first eight MOFA+ factors.

g Median genes versus median peaks for each donor. Point area is proportional to the number of retained nuclei. RNA and ATAC QC counts may differ slightly.

Fig2 | Genetic association and spatial neighbourhoods



a Colour shows S-LDSC tau z and point area shows nominal −log10 P, capped at 6. A black ring marks FDR < 0.05.

b Top 10 subtypes by mean RA S-LDSC statistic in the two European RA cohorts.

c Fine-mapped PIP mass relative to a permutation null. Shape and colour show the GWAS, point area shows nominal P and a black outline marks FDR < 0.05.

d Mean RA S-LDSC versus mean RA scDRS for states that have both estimates, with the Spearman correlation across states. M-6 has no S-LDSC estimate and is left out.

e Top variants, chosen by the highest supplied PIP with one variant per nearest protein-coding gene. Labels include the peak state. Grey dashes mark missing study records, not zero PIP. Exact variant IDs are in the source table saved with this figure.

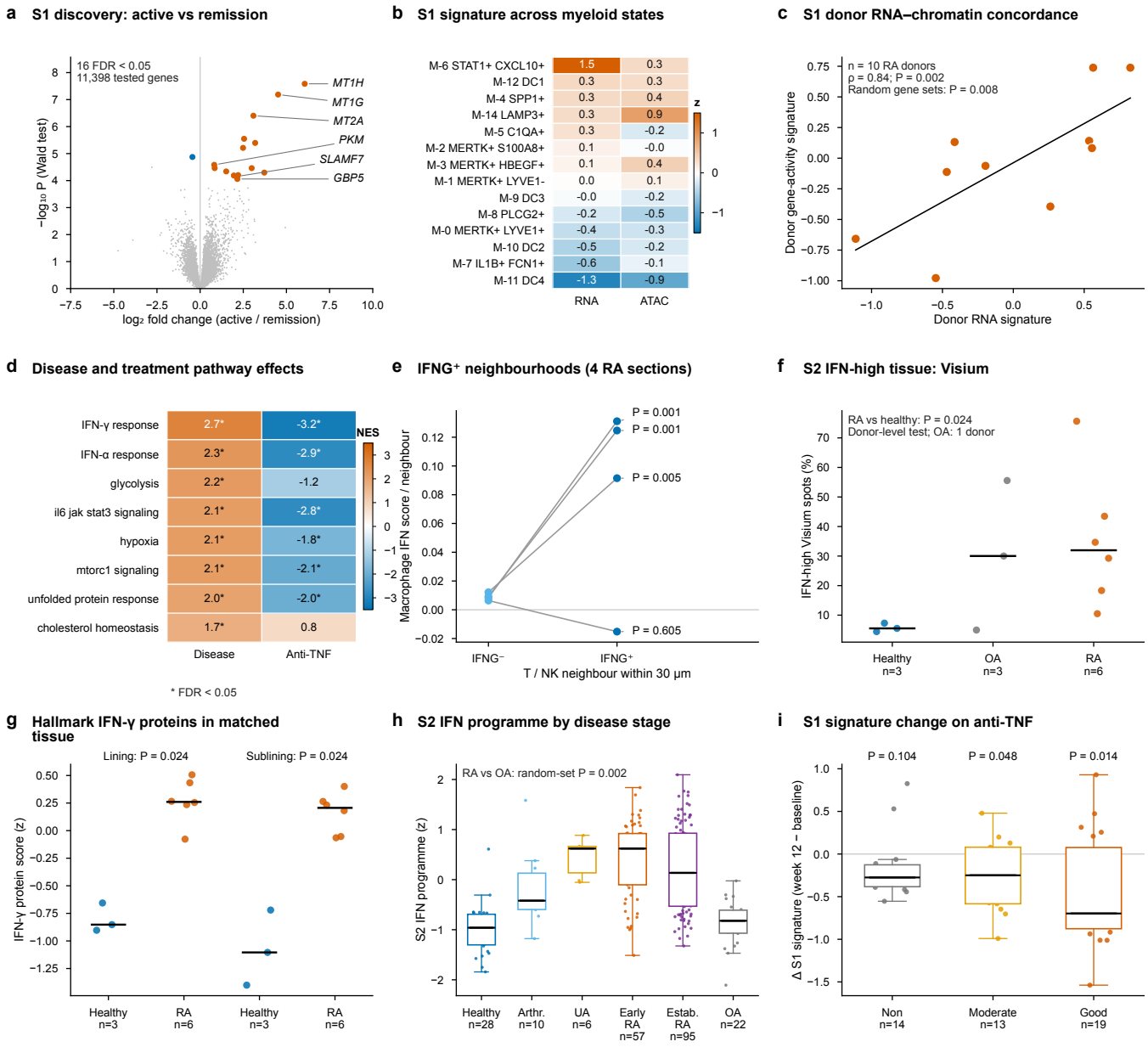
f Paired B-cell neighbourhood means within each section, with the positive/negative fold ratio. Lines join values from the same section.

g Xenium 5K section CT_21045-A-15, read from a processed plotting table made by step 20. The outlined square is the ROI shown in h. A cell counts as CXCL13-positive when its raw count is >0.

h 1.2-mm ROI centred on the densest 500-µm B-cell bin. It is chosen by a reproducible rule to show the structure of the aggregate, not to measure enrichment. Same cells and coordinates as g.

i Empirical permutation P for each section, capped at 1/1001, the smallest value possible with 1,000 permutations. Global label shuffling does not keep the spatial clustering of labels or the tissue compartments. These associations do not show causal organisation.

Fig3 | Separate discovery, IFN-programme and protein measurements



a DESeq2 active/remission effects and Wald statistics from the original analysis. P is recomputed from the stored Wald statistic so that no P value is rounded to zero. FDR status uses the supplied adjusted P.

b S1 is defined by 15 discovery genes. The 13 measurable genes give the RNA and ATAC gene-activity z scores shown for all 14 myeloid states, on one shared centred scale.

c S1 discovery-signature scores for each donor from RNA and ATAC gene activity (13 of 15 genes measurable), with the Spearman correlation and the supplied random-gene-set test. The line is a descriptive least-squares fit.

d The eight Hallmark pathways with the highest disease NES, with treatment NES for the same pathways. * marks supplied GSEA FDR < 0.05. Disease = active/remission macrophages; treatment = post/pre bulk in good responders.

e Paired regression coefficients for each section, for IFNG-positive and IFNG-negative T/NK neighbours. P values are one-sided spatial permutation tests from the existing table.

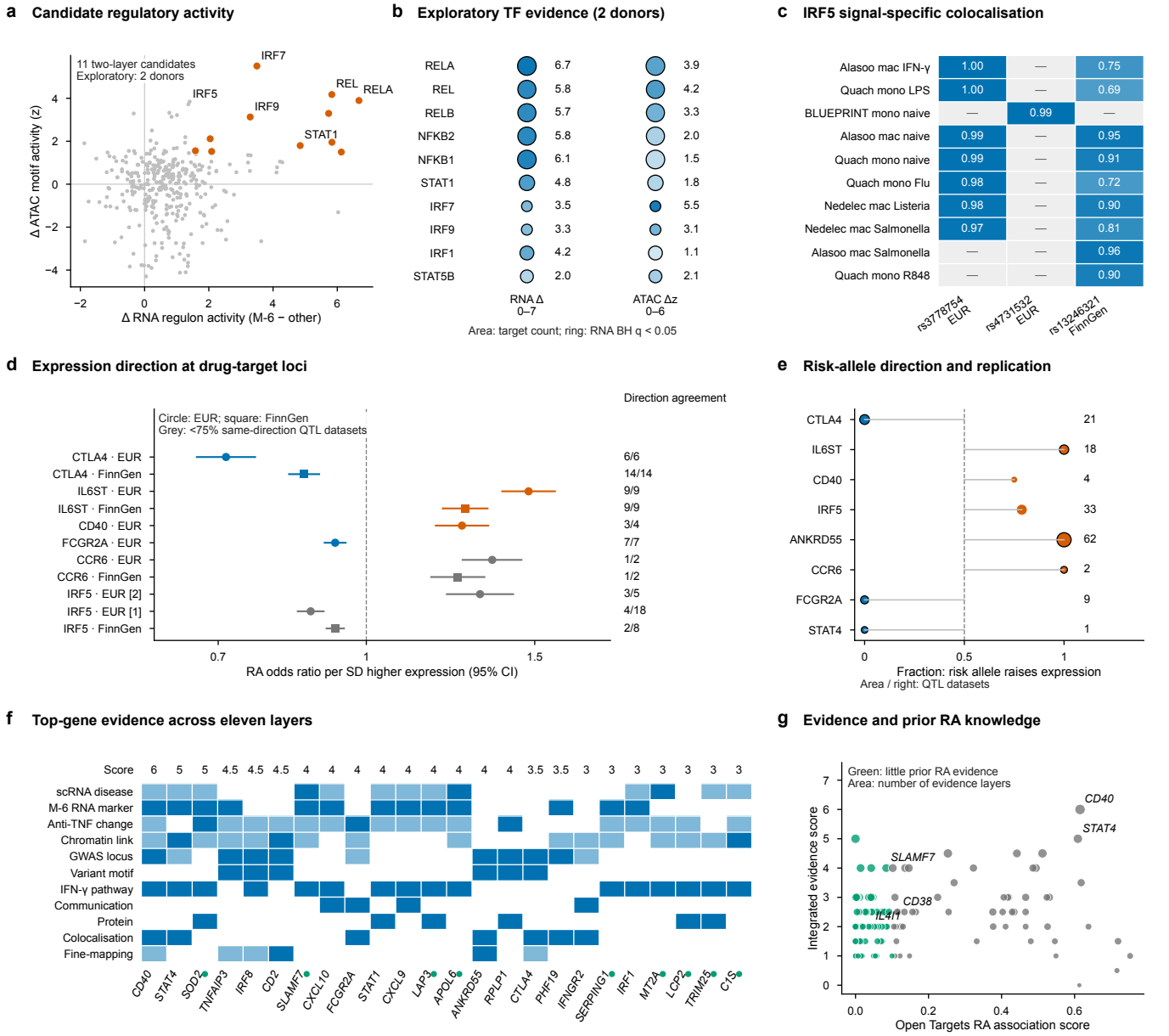
f S2 is the predefined 10-gene IFN programme. Each dot is a section and black bars are medians. The comparison shown is the supplied donor-level RA/healthy test, so section counts are not independent donor counts.

g Hallmark IFN-gamma protein score. Sublining joins immune_sublining, fibrous_sublining and sublining before the average within each donor, as in the statistical test (RA n=6, NH n=3 in each compartment). Black bars are medians. Two-sided Mann-Whitney P values are recomputed and checked against the source table. Protein-set specificity is supported in lining but not in sublining (random-set P=0.339).

h S2 is the predefined 10-gene IFN programme and is different from S1. Points are single biopsy scores. Boxes show the IQR and median, and whiskers 1.5×IQR. Counts are biopsies. The supplied random-gene-set comparison is shown.

i S1 is the 15-gene discovery signature (13 measured in this cohort). Points are the paired change for each patient after technical residualisation. Boxes show the IQR and median, and whiskers 1.5×IQR. The P values shown are supplied paired Wilcoxon tests, not comparisons between response groups.

Fig4 | Candidate regulators and expression-associated genetic evidence



a Differences in transcription-factor RNA regulon activity and chromVAR motif activity. Candidates pass BH q <0.05 across all tested RNA gene-label-permutation P values and rank in the top motif decile. Data come from two donors, and gene-label permutations are not donor replication. These are candidate regulators.

b Top 10 factors by summed RNA and motif rank. Point area = 15 + 2 sqrt(regulon target count). Each column has its own labelled colour scale in its own units (RNA 0–7, ATAC 0–6). A black outline marks RNA gene-label-permutation BH q <0.05. Printed numbers are unscaled effects.

c Highest IRF5 H4 for each project/biosample/condition and RA lead signal. The top 10 conditions by highest H4 are shown. Grey dashes mark missing records.

d Existing MR odds ratios and supplied 95% CI for six genes. All supplied estimates are shown, including more than one lead signal for some gene/GWAS pairs; numbers in brackets tell these apart, in source-table order. The right track gives same-direction/total datasets. Colour shows direction and grey marks agreement <75%. These are exploratory estimates, not measures of drug efficacy.

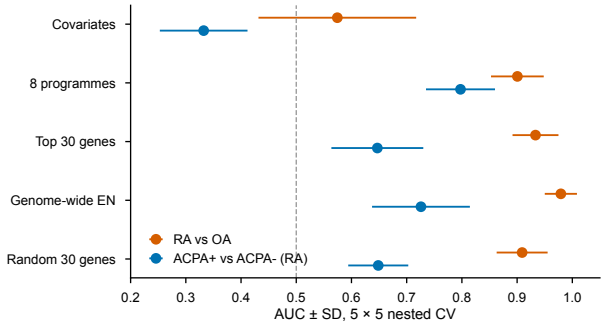
e Fraction of immune QTL datasets in which the risk allele raises expression. Point area and the number on the right give the number of datasets. A black ring marks the supplied consistency flag.

f Top 24 genes across the 11 original evidence layers, on a shared 0–1 scale. The original evidence score is printed above. White = 0, mid blue = 0.5, saturated blue = 1. A green marker below a gene shows low prior RA evidence, based on the supplied novelty label. The evidence layers are not assumed to be statistically independent.

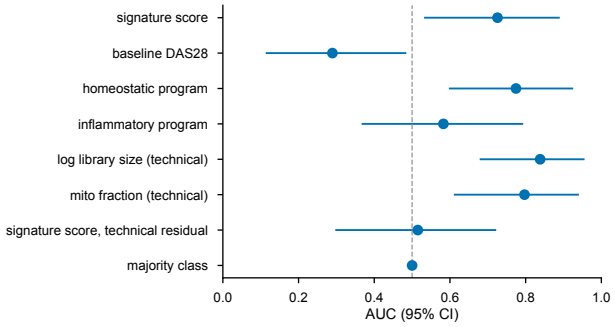
g All prioritised genes: original integrated score versus Open Targets association score. Point area = 5 + 2 × number of evidence layers. Green marks genes with the supplied low-prior-evidence label.

ExtData Fig1 | Baselines, robustness and boundary findings

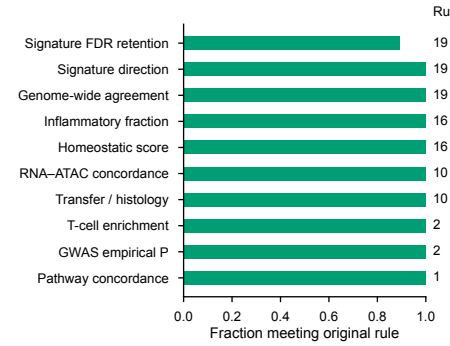
a Classification against matched baselines



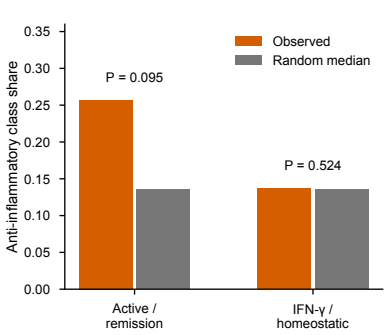
b Anti-TNF response prediction and confounding



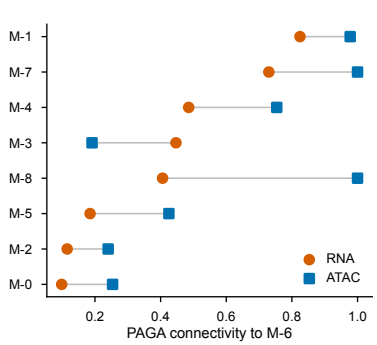
c Robustness across perturbations



d Drug-class reversal versus null



e M-6 graph neighbourhood



f Boundary findings and negative controls

Blood monocytes Paired signature test: P = 0.45; 13 pairs	**Bulk Tph score, adjusted for T-cell content** Random-set P = 0.81
Bulk deconvolution Failed the original sanity checks	**Myeloid S-LDSC** Negative conditional τ z in both RA GWAS
Active / remission classification AUC 0.50–0.60 in the original comparison	**Trajectory timing** Chromatin-before-RNA is not interpretable
RA / OA: prioritised versus random genes F2 vs random empirical p 0.2475	**East Asian T-cell peak enrichment** 1.26×; P = 0.22

a Mean AUC ± fold SD from the revised repeated nested CV, with all fitting done on training data only. Error bars are SD, not CI. For random controls, the error is the SD across the means of the draws. Observed and label-permuted F1 use the same 5 × 5 outer CV, 3-fold tuning and preprocessing. These are internal estimates from one cohort.

b AUC for good response versus nonresponse at baseline, with the supplied bootstrap 95% CI. The predictive signal seen at baseline is confounded by technical factors.

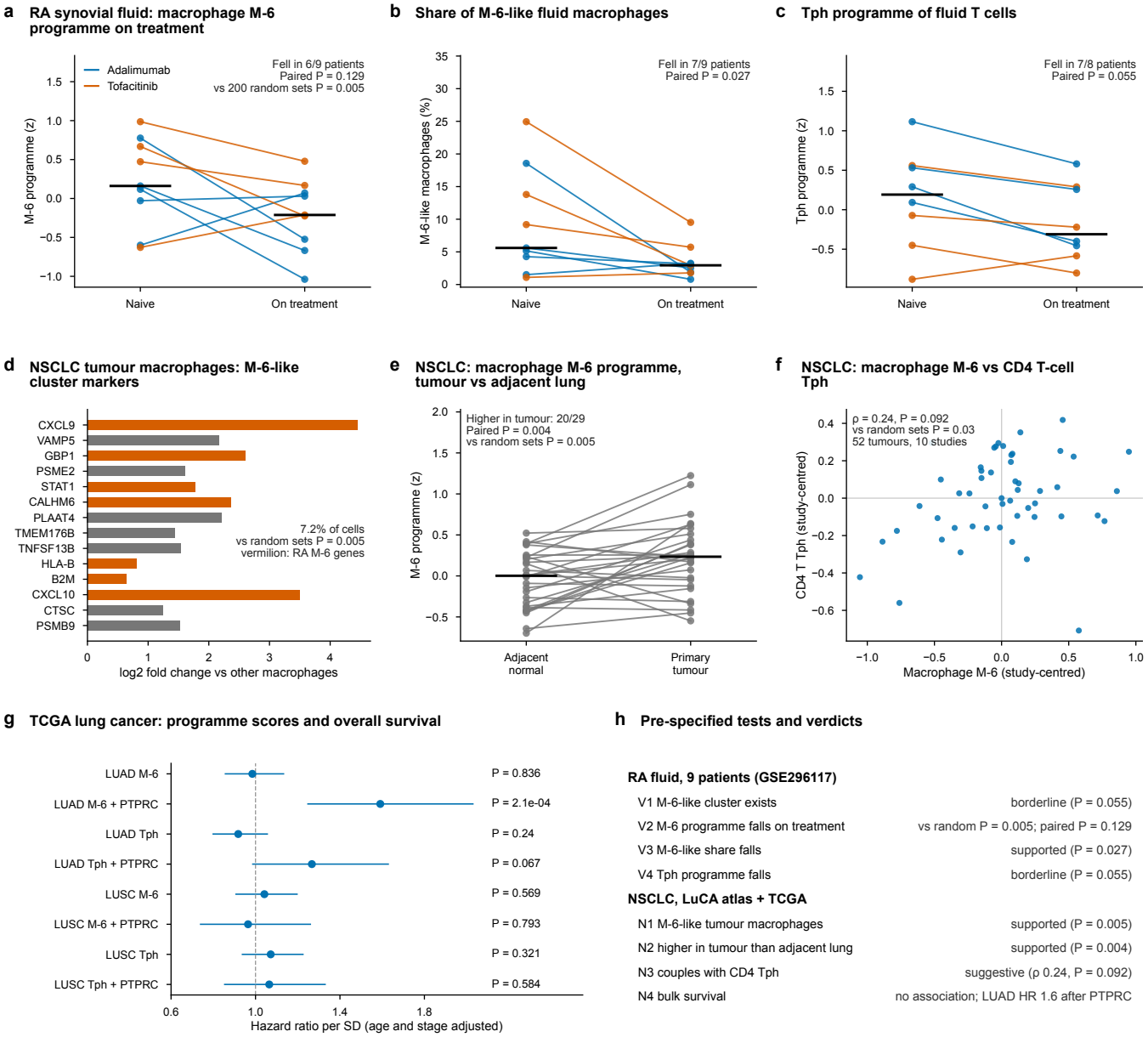
c Fraction of perturbations that meet each original rule. The right column gives the number of perturbations. Metric definitions and ranges are in robustness_summary.tsv.

d Observed share of anti-inflammatory compound classes versus the random median. P values are supplied empirical null comparisons.

e Existing RNA and ATAC PAGA connectivity to M-6. Closeness in the graph does not show developmental or temporal direction.

f Boundary findings. The blood P and the revised classifier random-set P are read from their result tables. The other entries come from the named source analyses.

ExtData Fig2 | Exploratory synovial-fluid and NSCLC extensions



h Pre-specified tests and verdicts

RA fluid, 9 patients (GSE296117)	
V1 M-6-like cluster exists	borderline (P = 0.055)
V2 M-6 programme falls on treatment	vs random P = 0.005; paired P = 0.129
V3 M-6-like share falls	supported (P = 0.027)
V4 Tph programme falls	borderline (P = 0.055)
NSCLC, LuCA atlas + TCGA	
N1 M-6-like tumour macrophages	supported (P = 0.005)
N2 higher in tumour than adjacent lung	supported (P = 0.004)
N3 couples with CD4 Tph	suggestive (ρ 0.24, P = 0.092)
N4 bulk survival	no association; LUAD HR 1.6 after PTPRC

a M-6 programme score (30 genes defined in RA synovial tissue) in synovial-fluid macrophage pseudobulk for each patient, treatment-naive and on treatment. Lines join the same patient, colour = drug and black bars are medians. The paired Wilcoxon test and the comparison with 200 expression-matched random gene sets come from ra_fluid_tests.tsv.

b Share of macrophages above the 90th percentile of treatment-naive M-6 cell scores, for each patient and time point.

c Tph programme score in T-cell pseudobulk. Eight patients have enough T cells at both time points.

d Top markers of the Leiden cluster with the highest M-6 score among 39,260 primary-tumour macrophages (LuCA core atlas, Harmony by study). Red marks genes in the RA M-6 programme. P compares this cluster with the best cluster of 200 random gene sets.

e Macrophage pseudobulk M-6 score for each donor in adjacent normal lung and primary tumour, for the 29 donors with >= 50 macrophages in both.

f Macrophage M-6 versus CD4 T-cell Tph programme score for each tumour, each centred within study. Spearman correlation on the study-adjusted residuals.

g Cox proportional hazards per SD of the bulk programme score in TCGA primary tumours, adjusted for age and stage, with and without adjustment for leukocyte abundance (PTPRC). Bars are 95% CI.

h Hypotheses as set in scripts/40_ra_fluid_treatment.py and 41_nsclc_extension.py before scoring. Verdicts use the stated two-sided tests.